

ML Correlation Model Integration — Walkthrough

What Was Built

A full ML-powered correlation analysis layer was added to ChainWatch, sitting between the individual risk agents and the LLM explanation step. Every real scan now feeds training data back to the model automatically.

New Files Created

File	Purpose
backend/services/data_logger.py	JSONL logger — records every real scan as training data
backend/ml/__init__.py	Package marker
backend/ml/seed_training_data.py	Generates 200 synthetic bootstrap training samples
backend/ml/correlation_model.py	Ridge Regression (risk score) + Random Forest (delay estimate)
backend/ml/model.pkl	Saved trained model (auto-reloaded on startup)
backend/data/risk_history.jsonl	Growing history of real + seed analysis runs
backend/agents/ml_agent.py	New agent that feeds real data to the ML model

Modified Files

File	Change
backend/models/schemas.py	Added MLAnalysisOutput schema + ml_analysis to SystemState
backend/orchestrator/orchestrator.py	6-step pipeline: added Step 4 (ML Agent) + data logging
backend/agents/aggregation_agent.py	Accepts ML score, blends 70% heuristic / 30% ML
backend/agents/explanation_agent.py	Passes ML data to LLM service
backend/services/llm_service.py	LLM prompt now includes ML score, delay, correlations, feature importance
backend/main.py	Risk-overview response includes ml_analysis, added /ml/status + /ml/retrain
frontend-next/lib/types.ts	Added MLAnalysis, MLCorrelationItem interfaces
frontend-next/components/RiskOverviewPanel.tsx	New ML comparison card in the dashboard
requirements.txt	Added scikit-learn>=1.4.0, pandas>=2.1.0

Pipeline (6 Steps)

1. News **Agent** → LLM classifies news → severity 1-5

2. Weather **Agent** → Threshold rules → severity 1-5

3. Port **Agent** → AIS vessel **count** → severity + heuristic delay

4. ML **Agent** → Ridge (risk **score**) + **Random** Forest (delay) + correlations ← NEW

5. Aggregation → 70% heuristic + 30% ML blended **score**

6. Explanation → GPT-4o-mini **with** ML context **in** prompt ← ENHANCED

Model Performance (on seed data)

Metric	Value
Risk Score R ²	1.0
Delay R ²	0.999
Training samples 200 (seed) + accumulating real data	
Confidence	100% at 200+ samples

Top Correlations Discovered

- vessel_count ↔ delay_hours: **0.936 (strong)**
- port_severity ↔ delay_hours: **0.922 (strong)**
- port_severity ↔ vessel_count: **0.905 (strong)**

Feature Importances (for delay prediction)

Feature	Importance
port_severity	64.7%
vessel_count	35.3%

How to Restart

Since you have major new Python files, restart the backend:

```
# In your uvicorn terminal — press Ctrl+C, then:
uvicorn backend.main:app --reload --port 8000
```

The frontend (npm run dev) needs no restart.

New API Endpoints

- GET /ml/status — Check if model is trained, sample count, R² scores
- POST /ml/retrain — Force retrain on all accumulated real data

Dashboard Changes

The **Full Risk Overview** panel now shows a new **ML CORRELATION ANALYSIS** card (indigo-themed) with:

- **Score Comparison:** Heuristic vs ML risk score side-by-side
- **Feature Importance:** Horizontal bars showing which factors drive the prediction
- **Top Correlations:** Cross-factor correlation scores with strength label
- **ML Delay Estimate:** ML-predicted delay alongside the heuristic delay

How Real Data Accumulates

Every time you run "Full Risk Overview" for a region:

1. The real AIS/weather/news values are logged to backend/data/risk_history.jsonl
2. After accumulating real runs, call POST /ml/retrain to retrain on real data
3. The model gradually shifts from seed data to your real observed patterns